Lab 07: Docker Containers and Images

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Section: A

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Task 1: Install Docker and Docker Compose

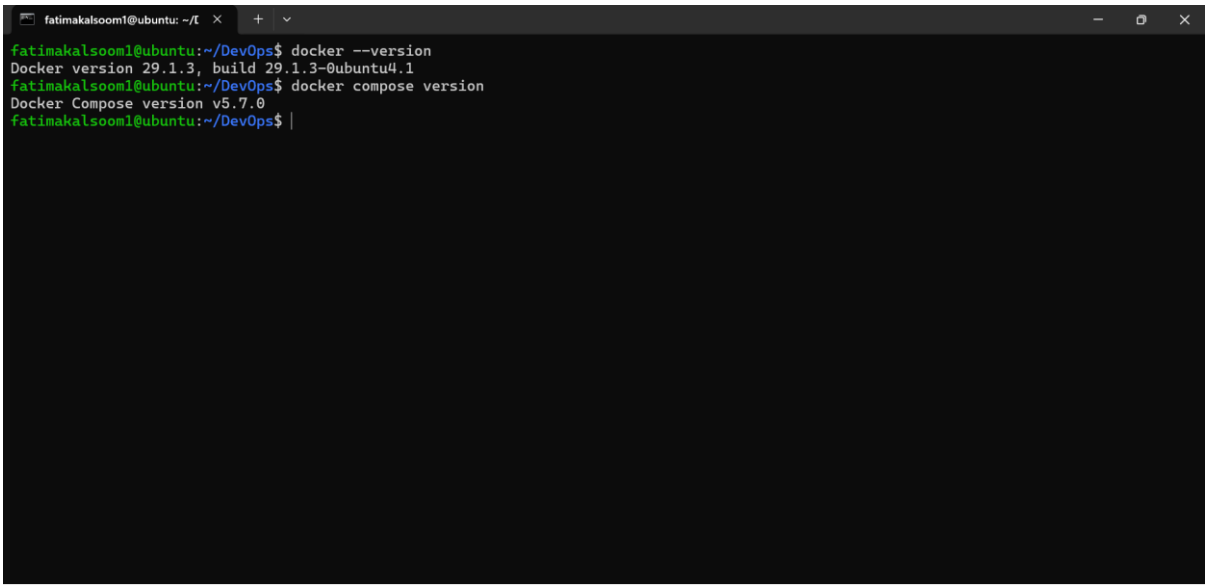


Figure 1: Docker and Docker Compose version check

Task 2: Run Two PostgreSQL Versions in Docker

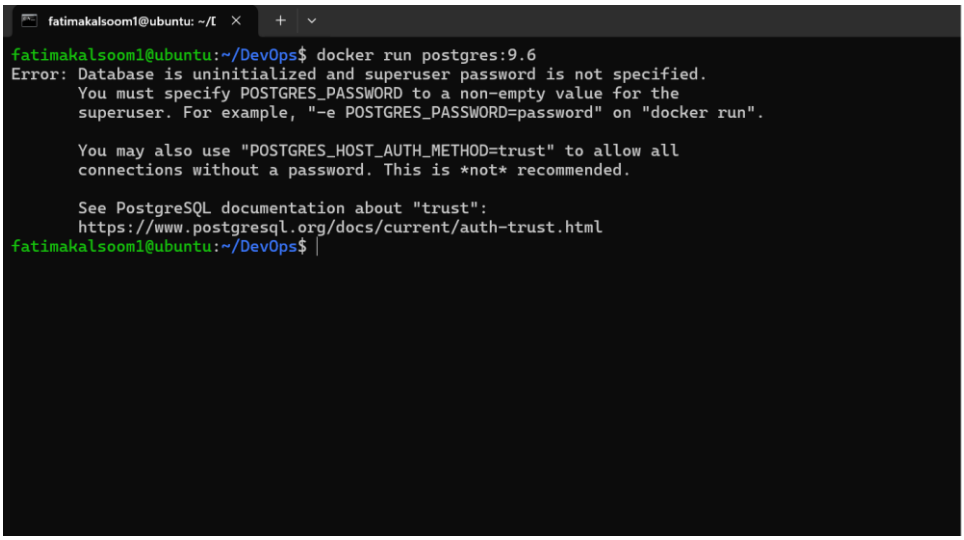


Figure 2: PostgreSQL 9.6 container exits with the missing POSTGRES_PASSWORD error

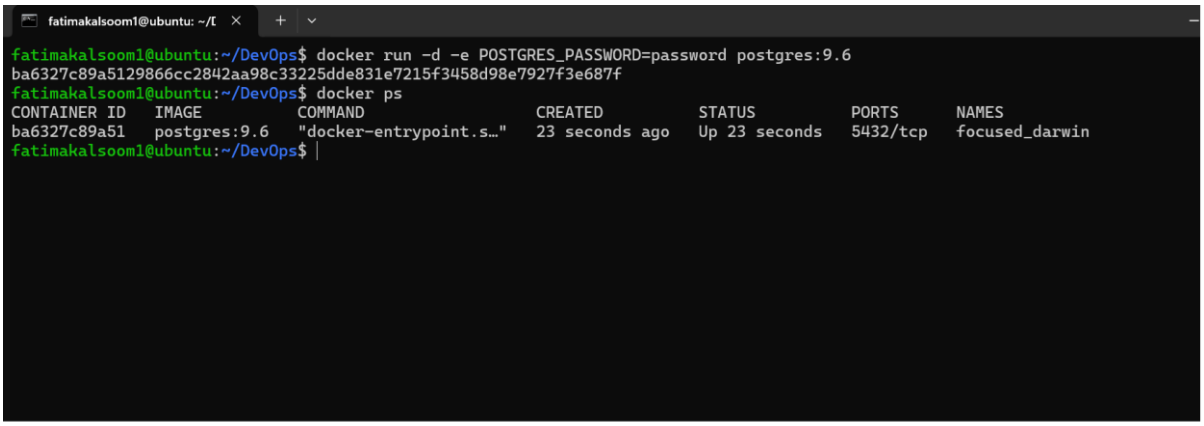


Figure 3: PostgreSQL 9.6 container running with the password set

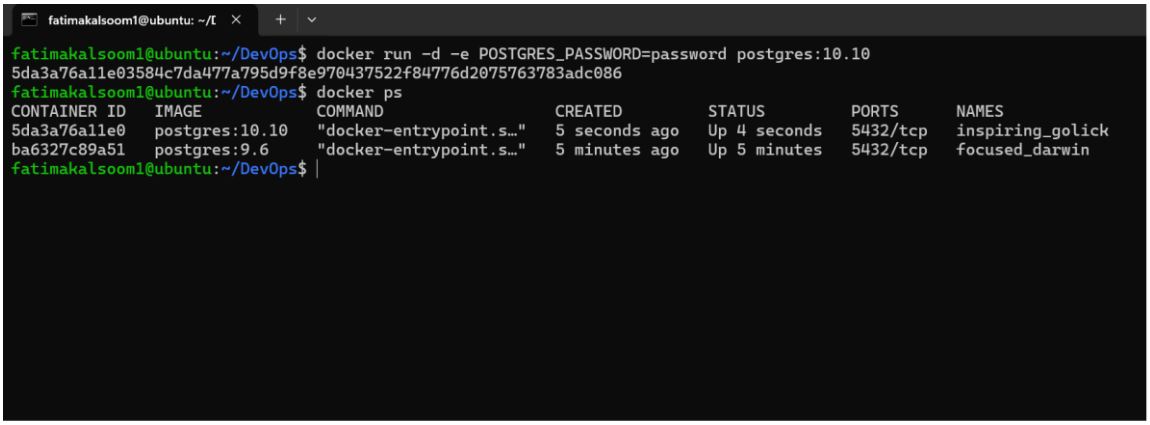


Figure 4: PostgreSQL 9.6 and 10.10 containers running together

Task 3: Inspect Docker and Manage Containers and Images

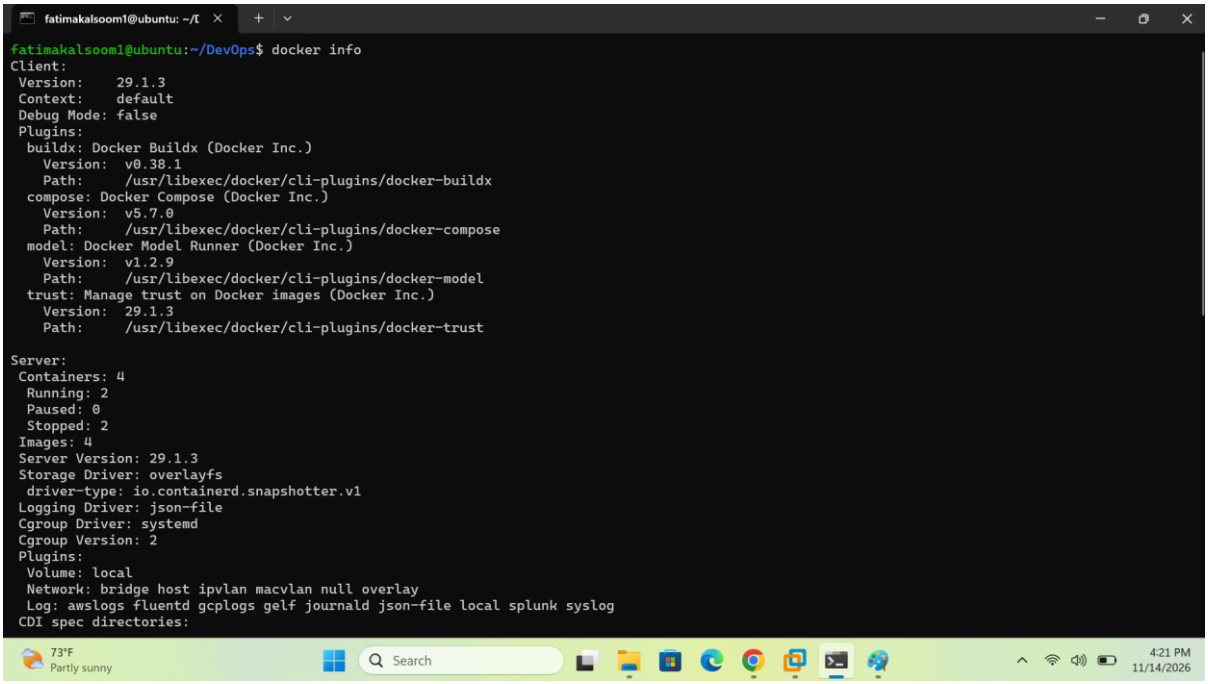


Figure 5: Docker installation details using docker info

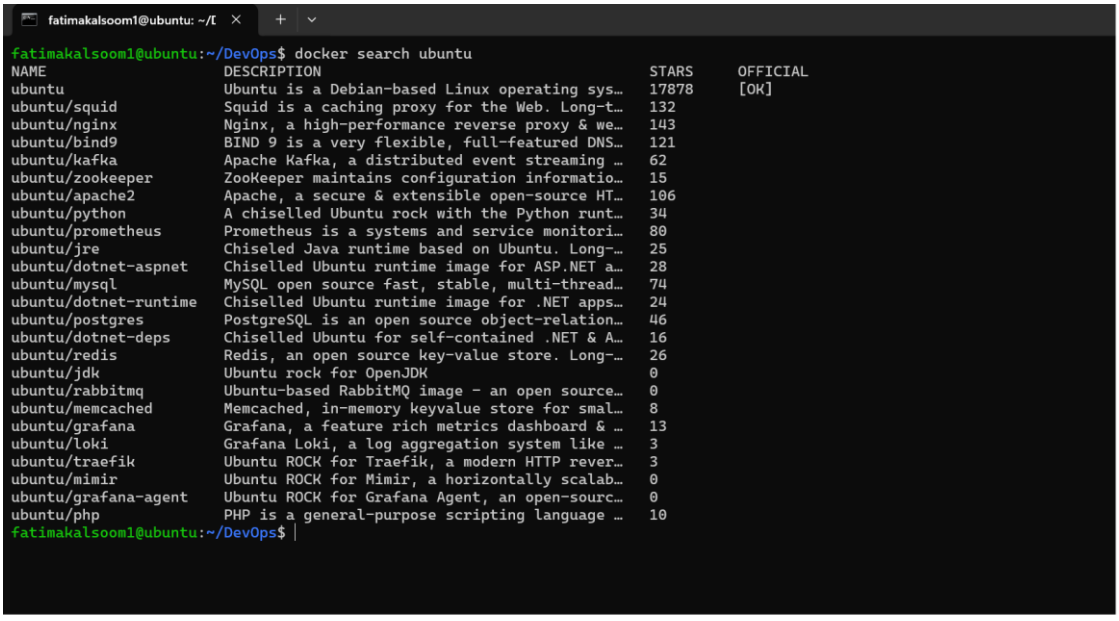


Figure 6: Searching Docker Hub for Ubuntu images

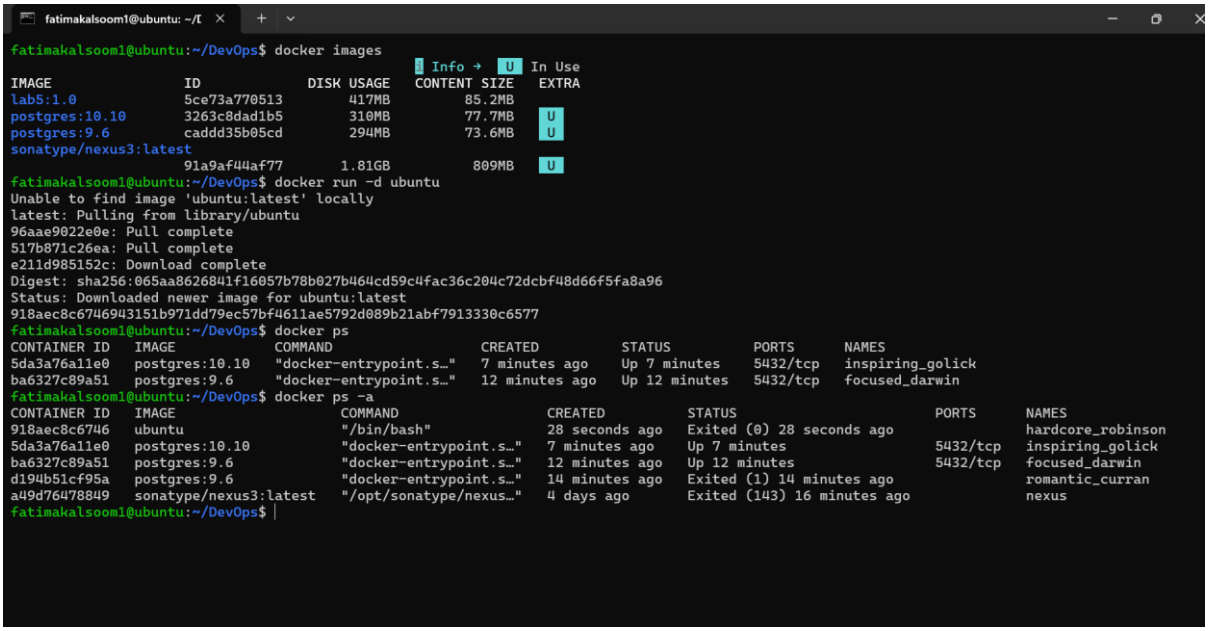


Figure 7: Ubuntu container shown as Exited in docker ps -a

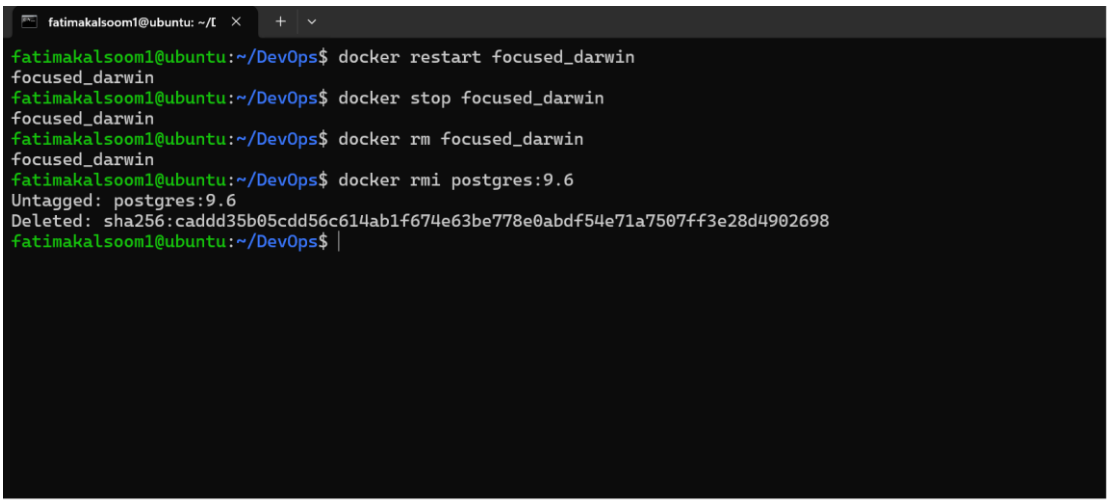


Figure 8: Restarting, stopping and removing the PostgreSQL 9.6 container and its image

Task 4: Run Two NGINX Containers on Different Ports

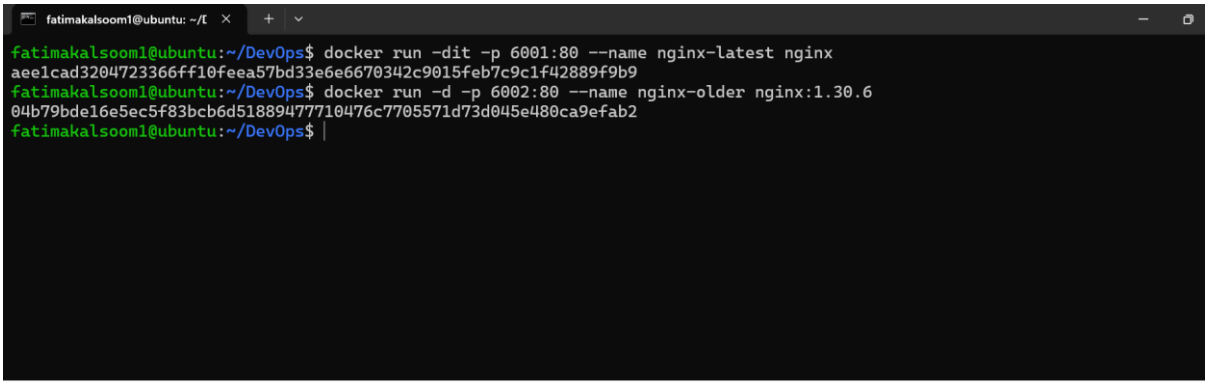


Figure 9: Starting the nginx-latest and nginx-older containers

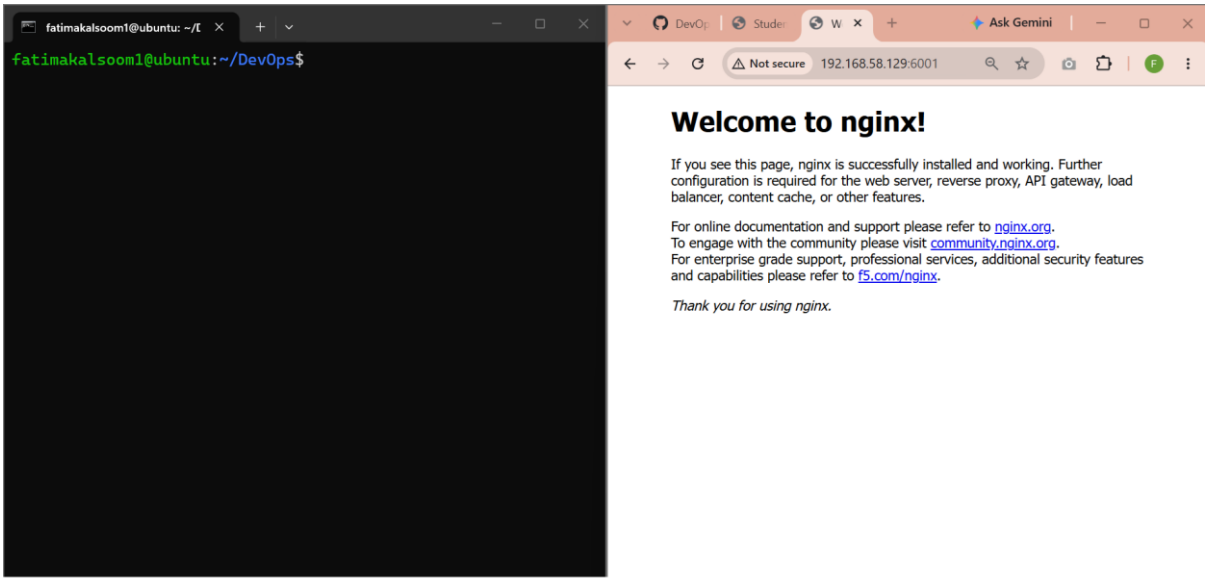


Figure 10: NGINX welcome page of nginx-latest on port 6001

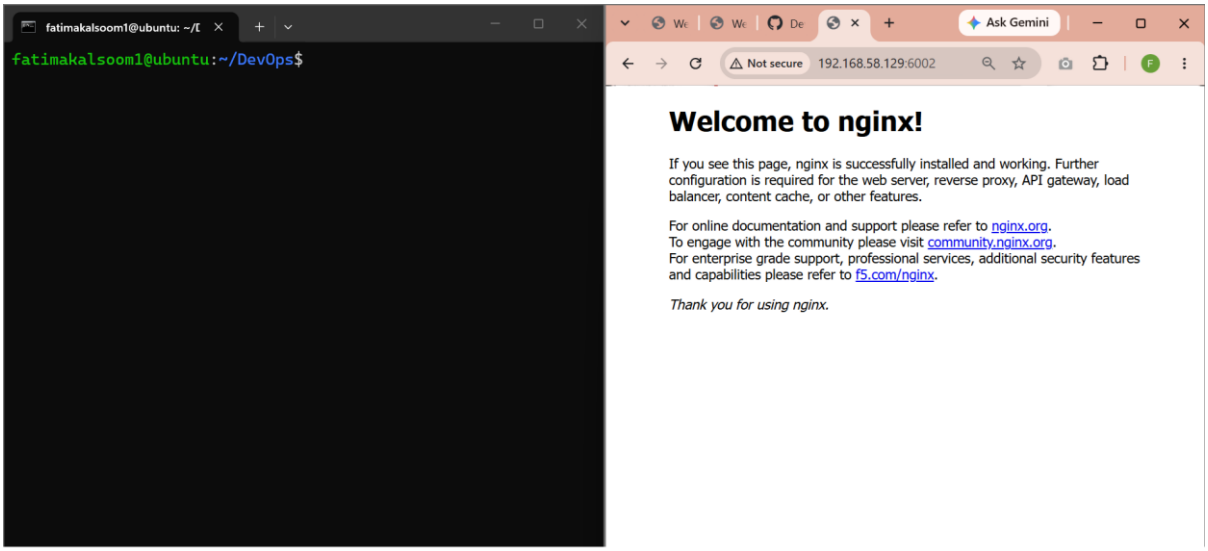


Figure 11: NGINX welcome page of nginx-older on port 6002

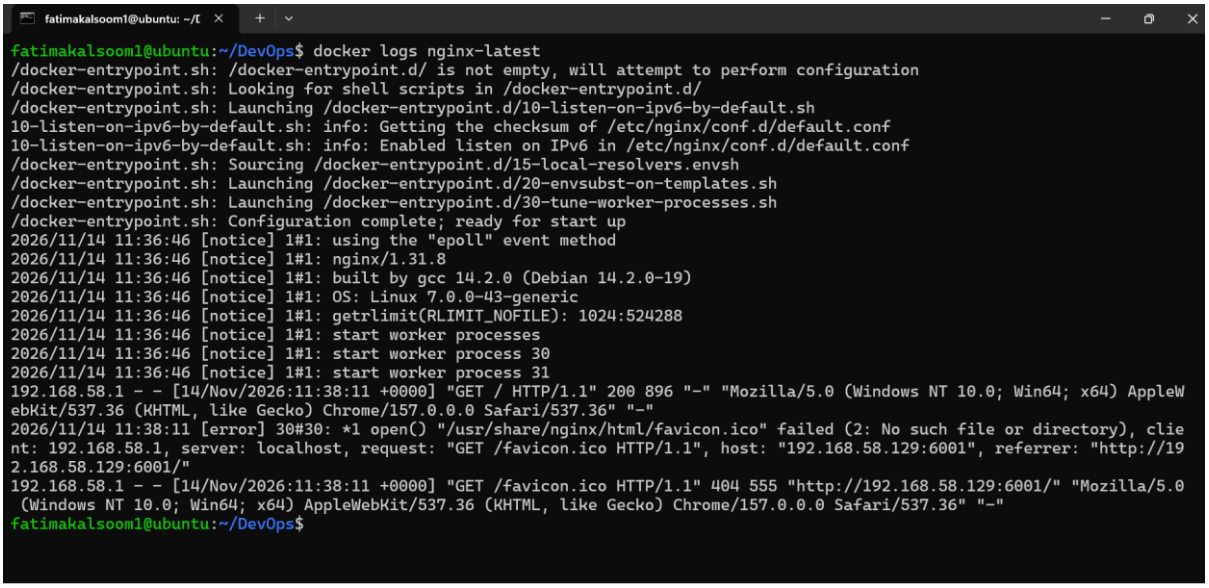


Figure 12: Container logs of nginx-latest showing the browser request

Task 5: Work Inside a Container and Save a New Image

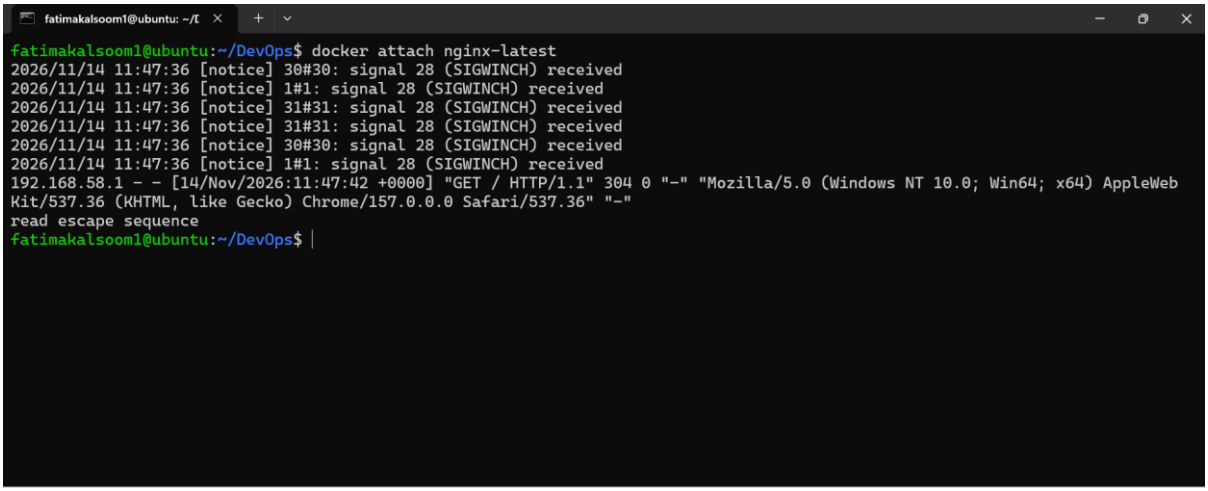


Figure 13: Attaching to and detaching from the nginx-latest container

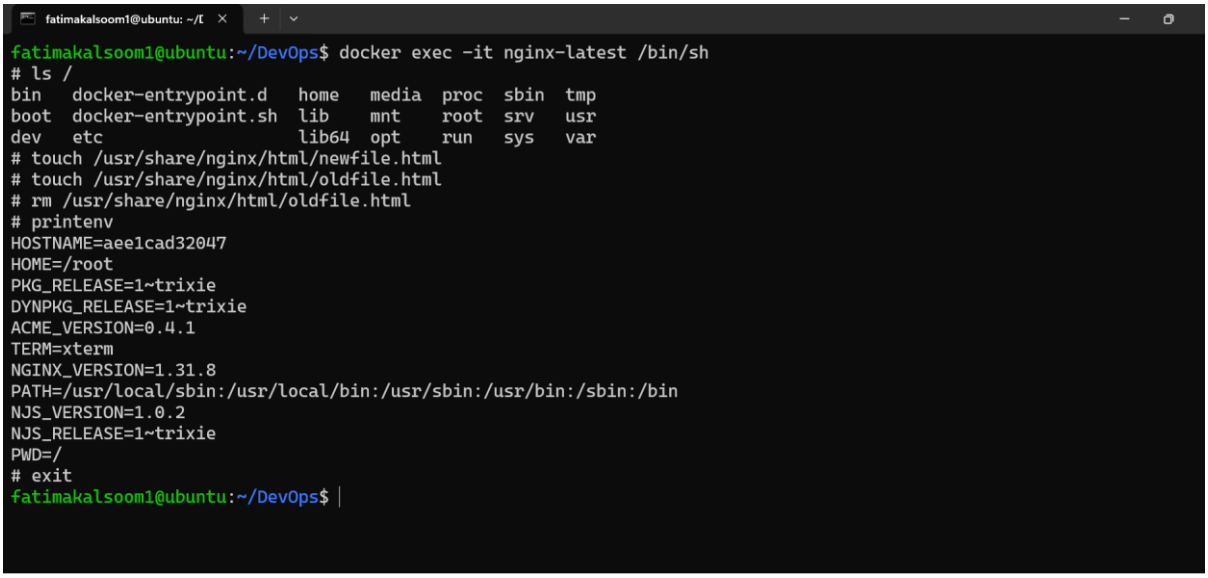


Figure 14: Running commands in a separate shell inside the container

```
fatimakalsoom1@ubuntu: ~/[
fatimakalsoom1@ubuntu:~/DevOps$ docker diff nginx-latest
C /etc
C /etc/nginx
C /etc/nginx/conf.d
C /etc/nginx/conf.d/default.conf
C /run
A /run/nginx.pid
C /usr
C /usr/share
C /usr/share/nginx
C /usr/share/nginx/html
A /usr/share/nginx/html/newfile.html
C /var
C /var/cache
C /var/cache/nginx
A /var/cache/nginx/client_temp
A /var/cache/nginx/fastcgi_temp
A /var/cache/nginx/proxy_temp
A /var/cache/nginx/scgi_temp
A /var/cache/nginx/uwsgi_temp
fatimakalsoom1@ubuntu:~/DevOps$
```

Figure 15: Filesystem changes shown by docker diff

```
fatimakalsoom1@ubuntu: ~/[
fatimakalsoom1@ubuntu:~/DevOps$ docker inspect nginx-latest
[
    {
        "Id": "aee1cad3204723366ff10feea57bd33e6e6670342c9015feb7c9c1f42889f9b9",
        "Created": "2026-11-14T11:36:46.426118301Z",
        "Path": "/docker-entrypoint.sh",
        "Args": [
            "nginx",
            "-g",
            "daemon off;"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 5021,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-11-14T11:36:46.491402319Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
        "Image": "sha256:165ddde8751e75a1c907d24ef10d9f6deb08b57e0fe56caf67512486aff76d82",
        "ResolvConfPath": "/var/lib/docker/containers/aee1cad3204723366ff10feea57bd33e6e6670342c9015feb7c9c1f42889f9b9/resol
v.conf",
        "HostnamePath": "/var/lib/docker/containers/aee1cad3204723366ff10feea57bd33e6e6670342c9015feb7c9c1f42889f9b9/hostnam
e",
        "HostsPath": "/var/lib/docker/containers/aee1cad3204723366ff10feea57bd33e6e6670342c9015feb7c9c1f42889f9b9/hosts",
        "LogPath": "/var/lib/docker/containers/aee1cad3204723366ff10feea57bd33e6e6670342c9015feb7c9c1f42889f9b9/aee1cad32047
72°F
Mostly sunny
Search
4:51 PM
11/14/2026
```

Figure 16: Container state and configuration using docker inspect

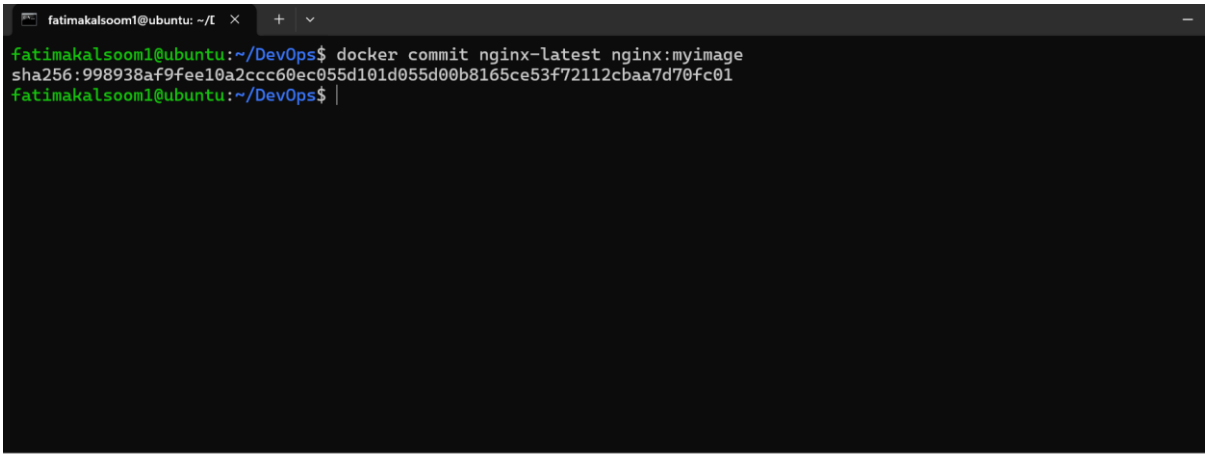


Figure 17: Saving the container as the new image nginx:myimage